

A close-up, low-angle shot of a field of green grass with seed heads. The grass is blowing vigorously in the wind, creating a sense of motion and texture. The seed heads are prominent, showing their feathery structure. The background is a soft-focus expanse of more grass.

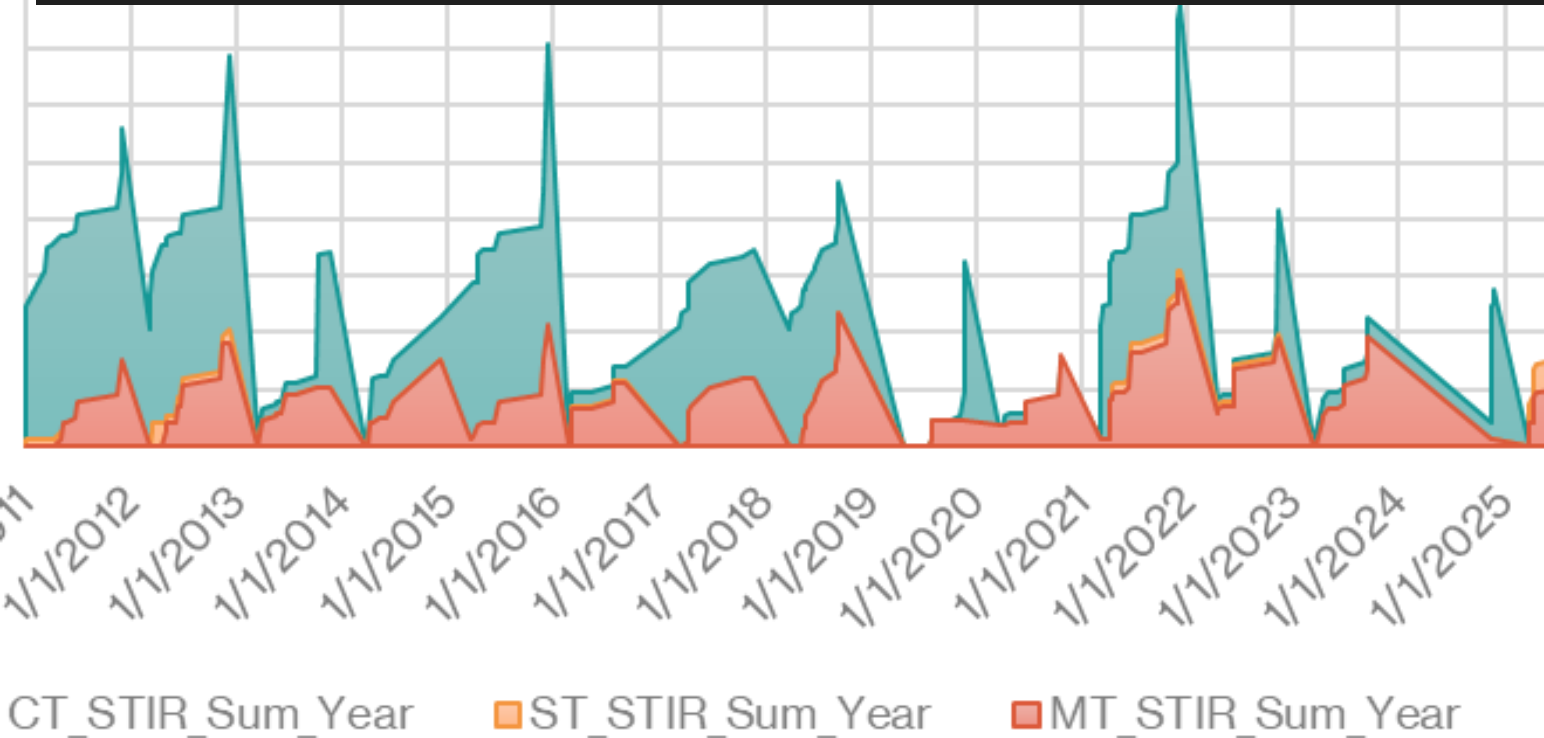
Disentangling Tillage Effects And Temporal Persistence In Edge-of-field Water Quality Using A Bayesian Causal Framework

A.J. Brown

20 Feb 2026

Annual STIR Accumulation

$$\mu_i = \alpha_a + \beta_a^{(STIR)} S_i + \beta_a^{(CIN)} CIN_i + \beta^{(VOL)} V_i + \beta^{(IRR)} IRR_i + \beta_a^{(DUP)} DUP_i + \gamma_{a,B_i}^{(B)} + \gamma_{a,S_i}^{(S)} + \gamma_{a,F_i}^{(F)} + f_{a,Y_i}$$



The Fork

$X \leftarrow Z \rightarrow Y$

The Pipe

$X \rightarrow Z \rightarrow Y$

The Collider

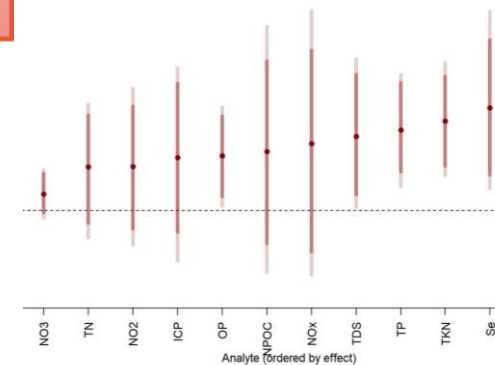
$X \rightarrow Z \leftarrow Y$

The Descendant

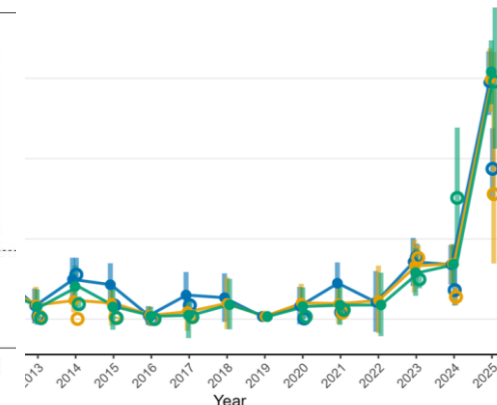
$X \rightarrow Z \rightarrow Y$



Analyte-specific STIR effects with 89% & 95% CIs

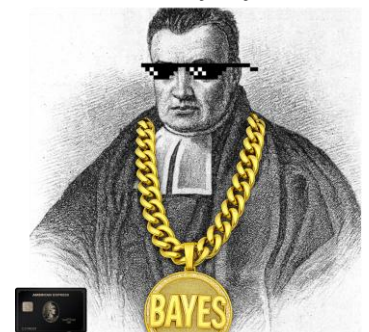


ual load by treatment



Previously on
“AJ’s Tillage Adventures”

“Full Luxury Bayes”



The Empire Strikes Back (With Model Improvements)



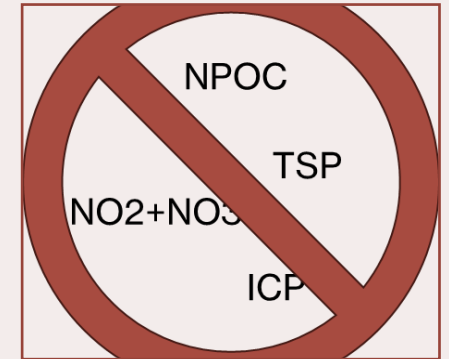
Crop Type



Residue



Irrigation Count
by Analyte



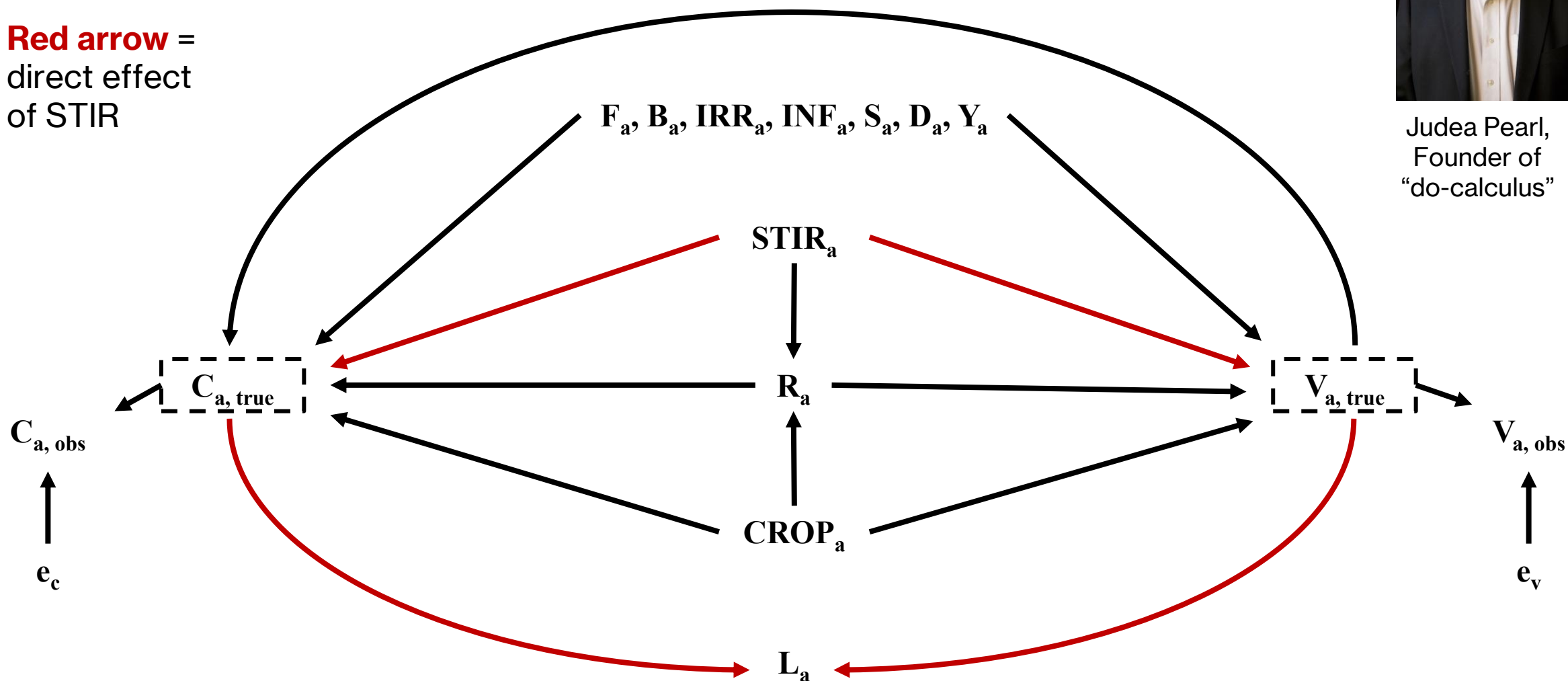
Reduced
Analyte Types

The Good, the Bad, and the Confounded



Judea Pearl,
Founder of
“do-calculus”

Red arrow =
direct effect
of STIR



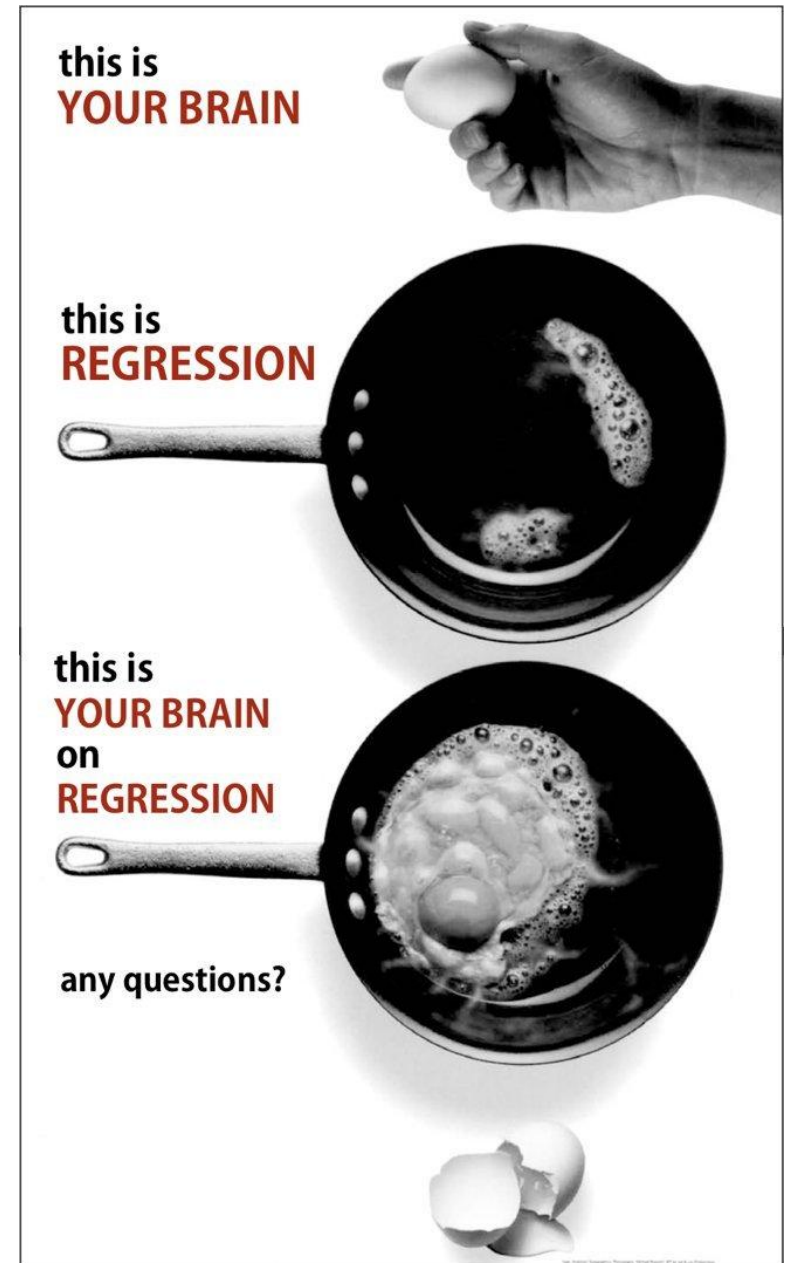
Fast and the Furious 2: Hamiltonian Drift

Think like a graph! Not a Regression!

DAG says we need 4 models running simultaneously with Hamiltonian Monte Carlo sampling

1. **Concentration model (multi-analyte, latent true concentration with GP structure)**
2. **Volume model (latent true volume)**
3. **Residue model (logit-normal regression with missing-data imputation)**
4. **Inflow concentration model (missing-data imputation)**

All models create a single hierarchical generalized linear mixed model with latent states and Gaussian process structure.



RESULTS 2: JUDGEMENT DAY

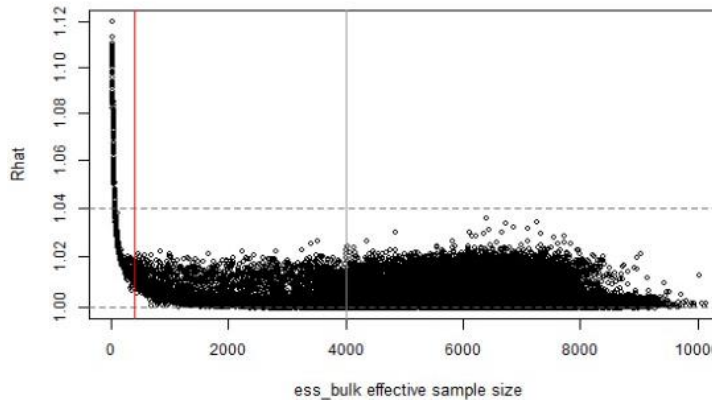
By grounding inference in formal probability and causal structure, we ensure the mathematics is internally consistent. When conclusions diverge from expectations, the assumptions, not the calculus, deserve scrutiny.

“It may not be the results you want, but it’s the results you need. Trust the process.” – Richard McElreath



AWQP's Next Top Model

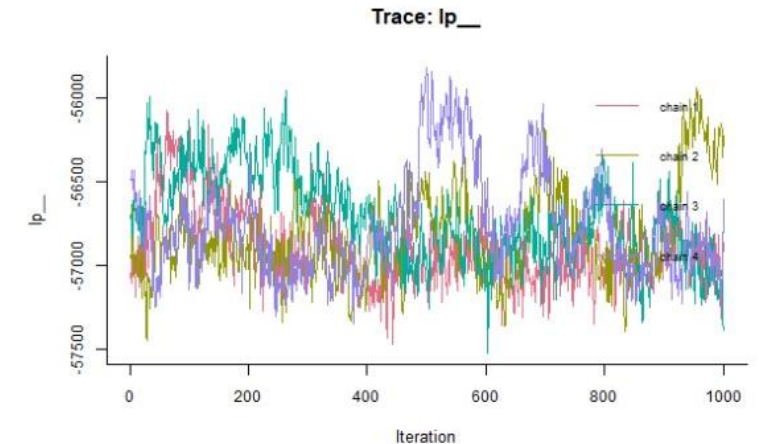
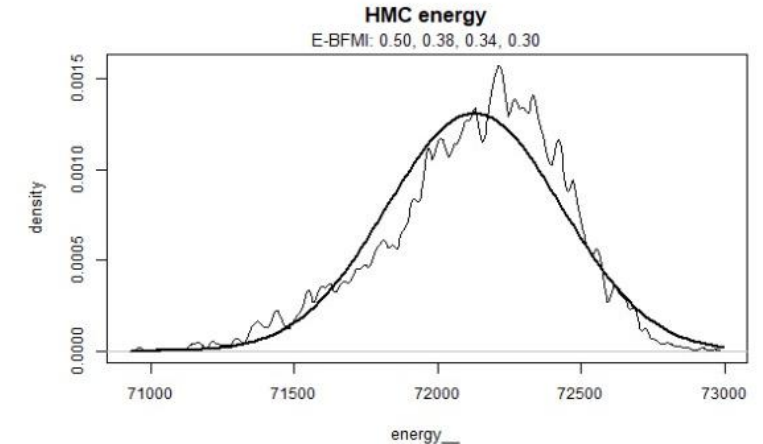
- Degrees of freedom? - Meet Bayes
 - **139,793** parameters estimated across **10,804** analyte observations
 - ~2,236,688,000 posterior values generated (~5.5 hours)
 - Includes latent “true” states, missing-data imputation, and posterior predictive quantities
- Model convergence – good!



Sampler issues

31

Divergent transitions



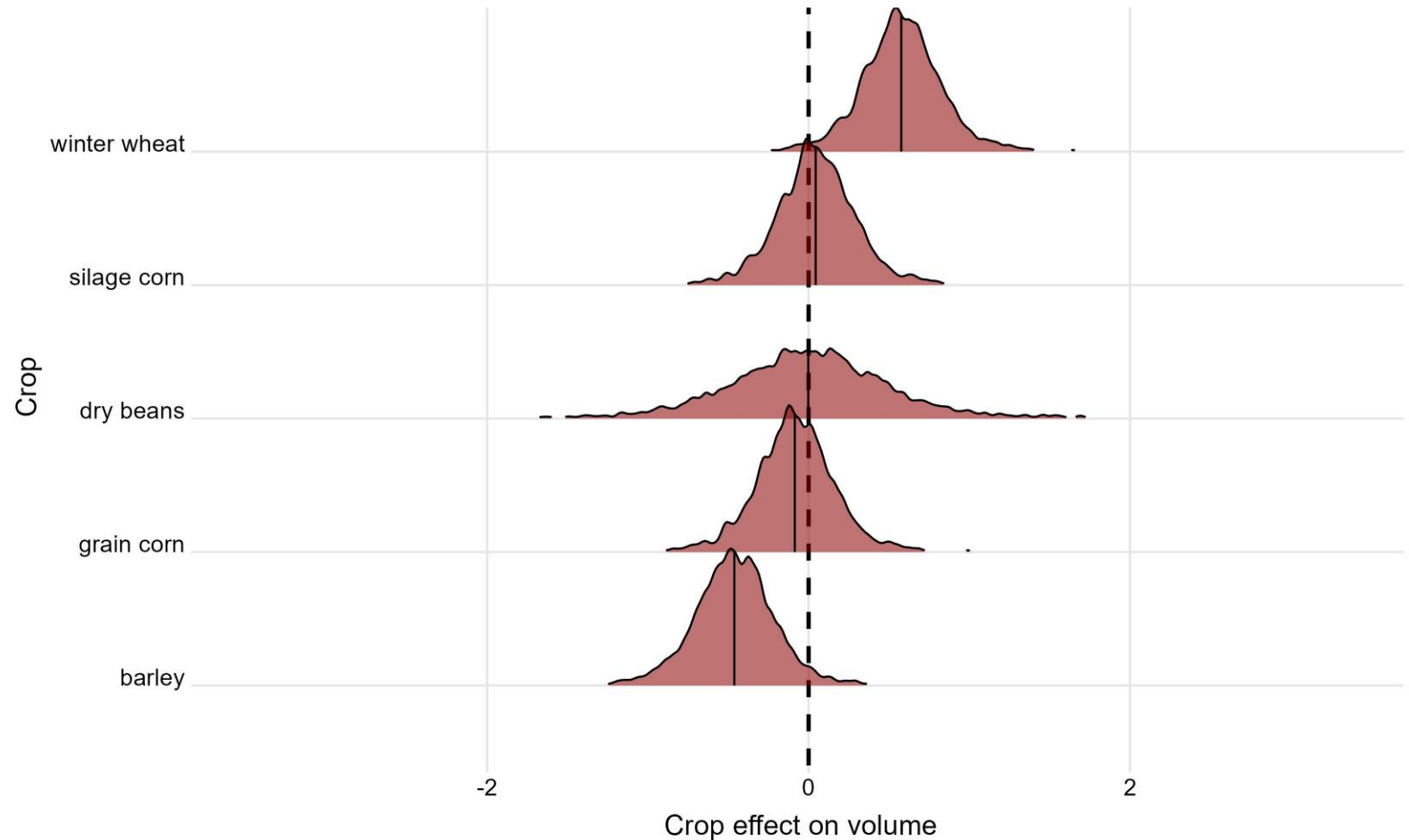
Children of the Corn (Crop Effects on Conc.)

If a ridge overlaps zero, it does not mean “no effect.”

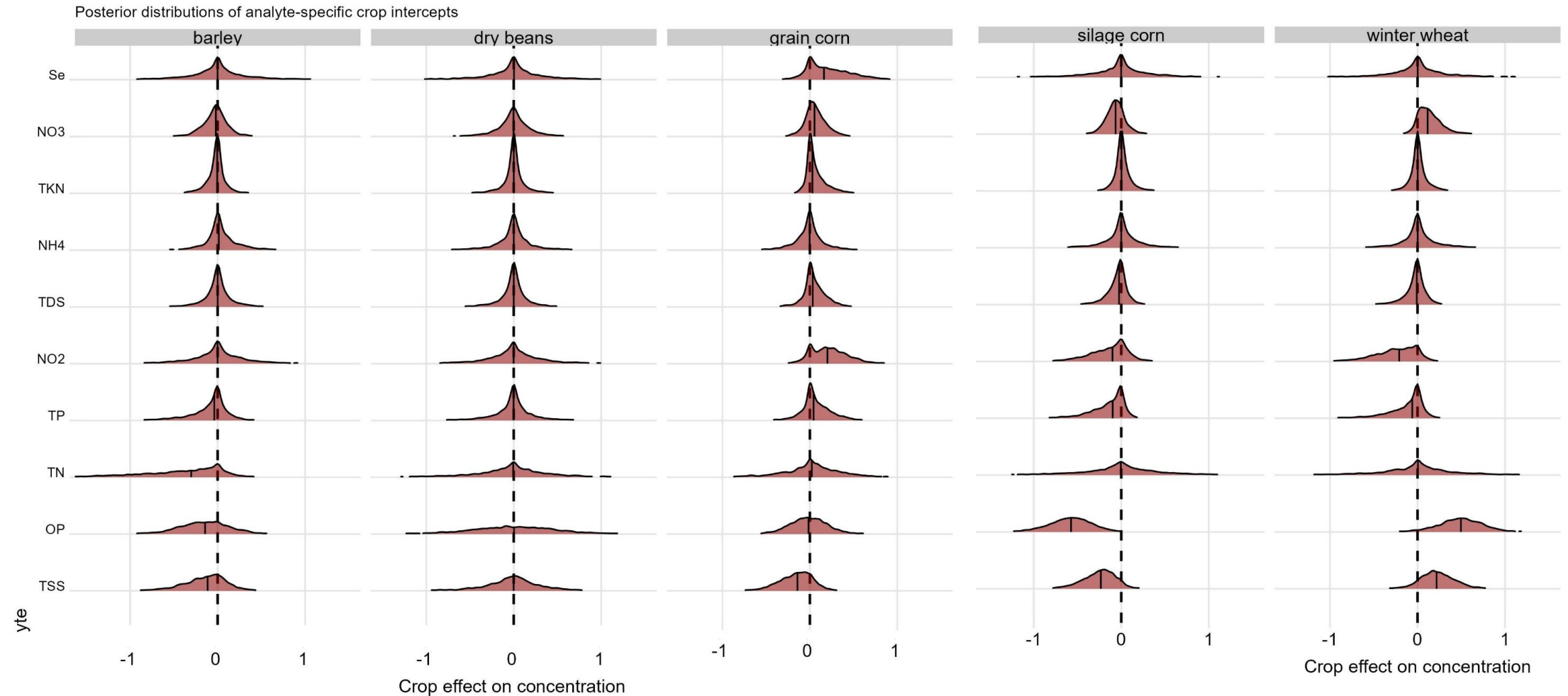
It means the model assigns some probability to both positive and negative values.

These are called **ridgeline plots**.

- Each ridge shows the distribution of plausible effect sizes.
- The solid vertical bar in each ridge is the mean.
- Width shows uncertainty.
- Direction and overlap tell you how confident we are in differences.

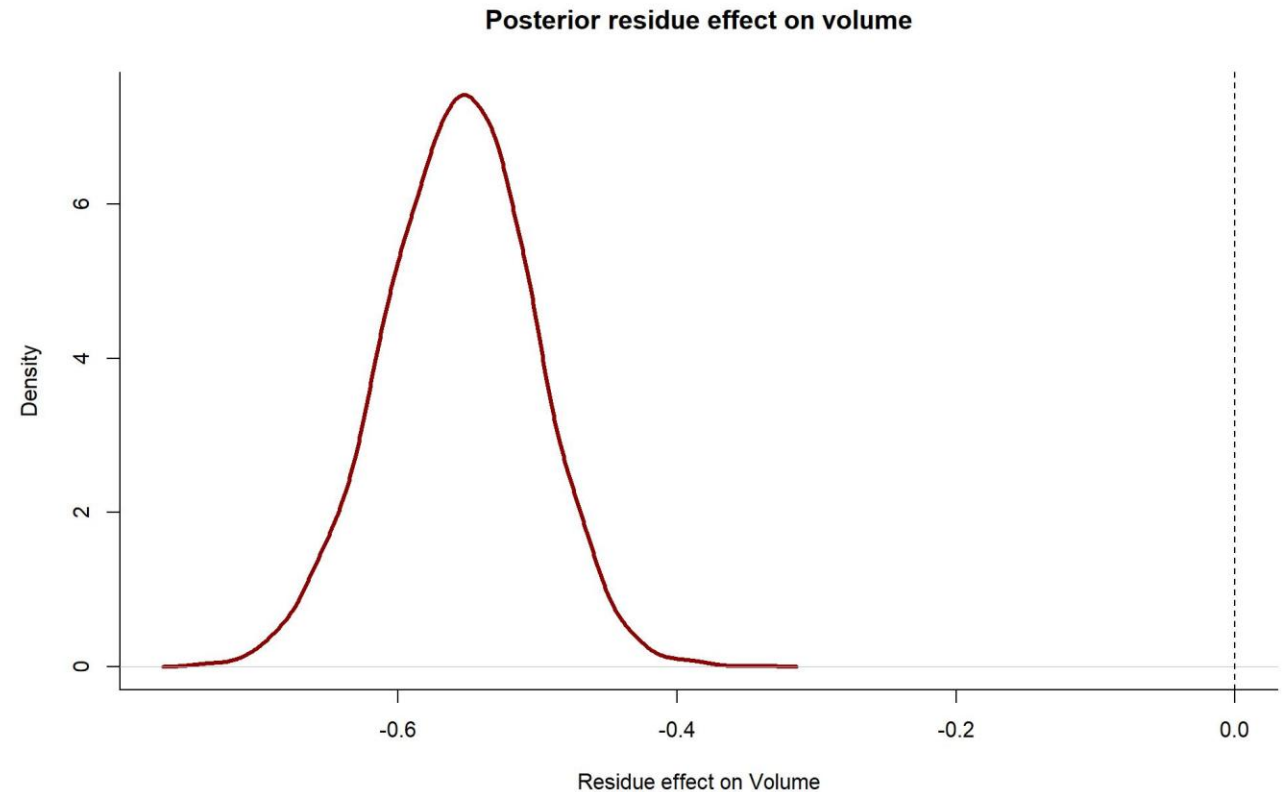
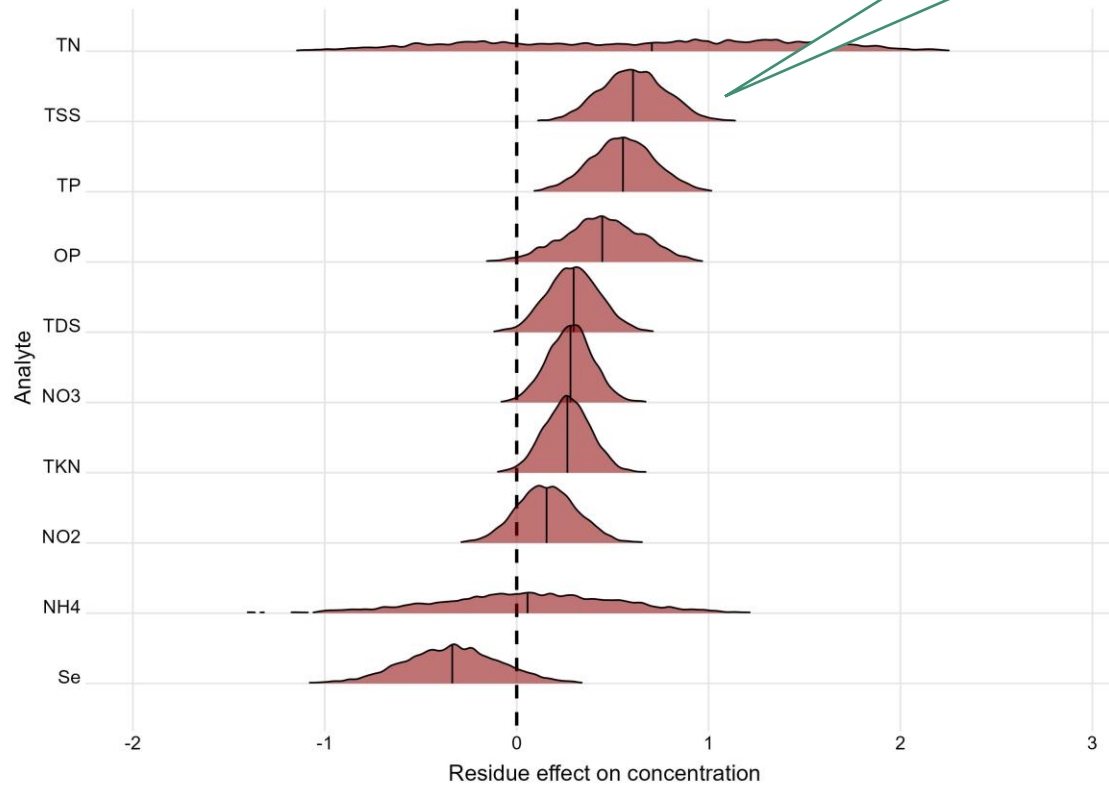


Crop Effects on Concentration

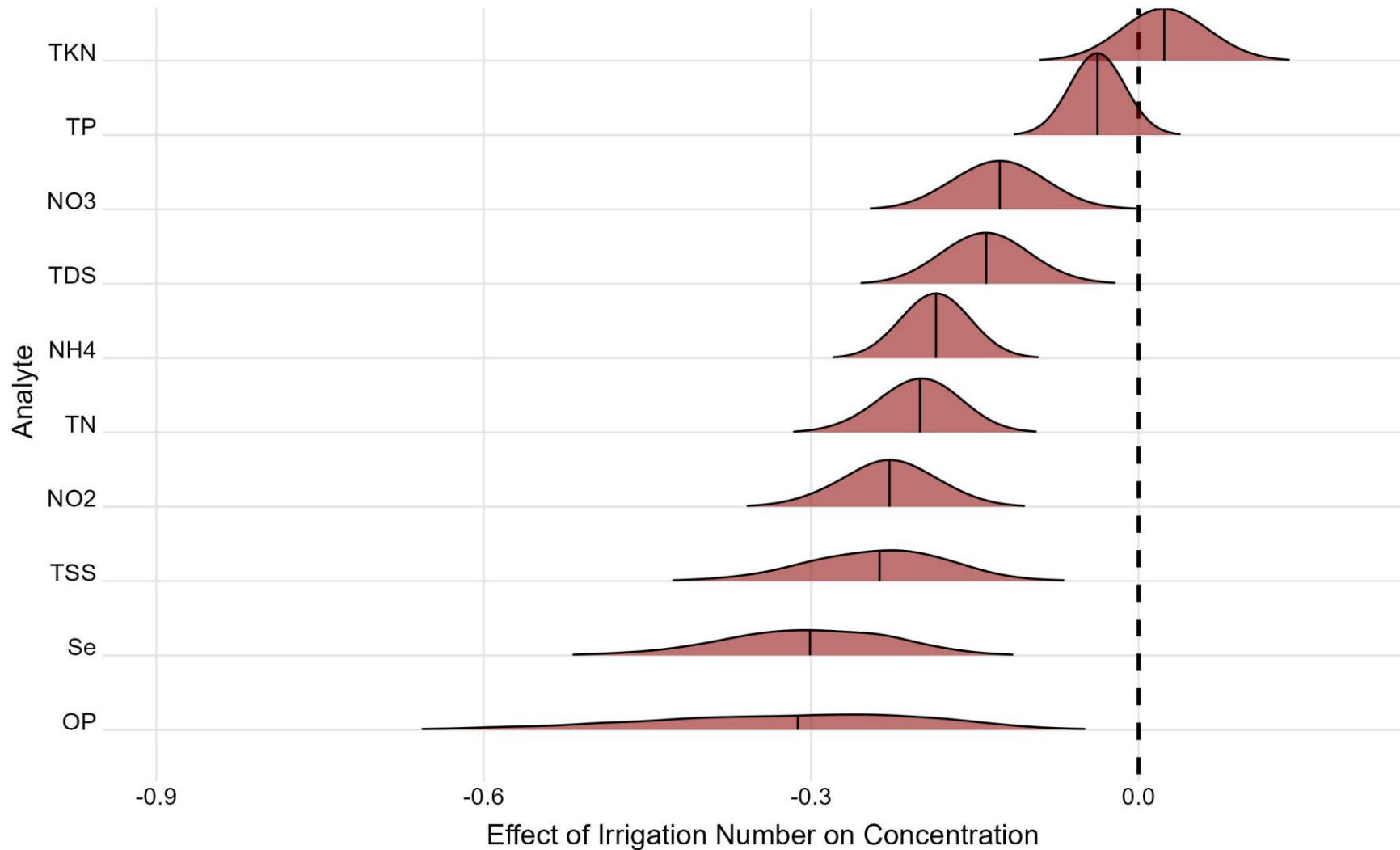


Residue Impacts

- Holding everything else constant
- Recall, this is residue “at planting”, not at the time of irrigation
- Fields with higher residue often get a furrow clean before Irr. 1
- Residue itself flows down the furrow

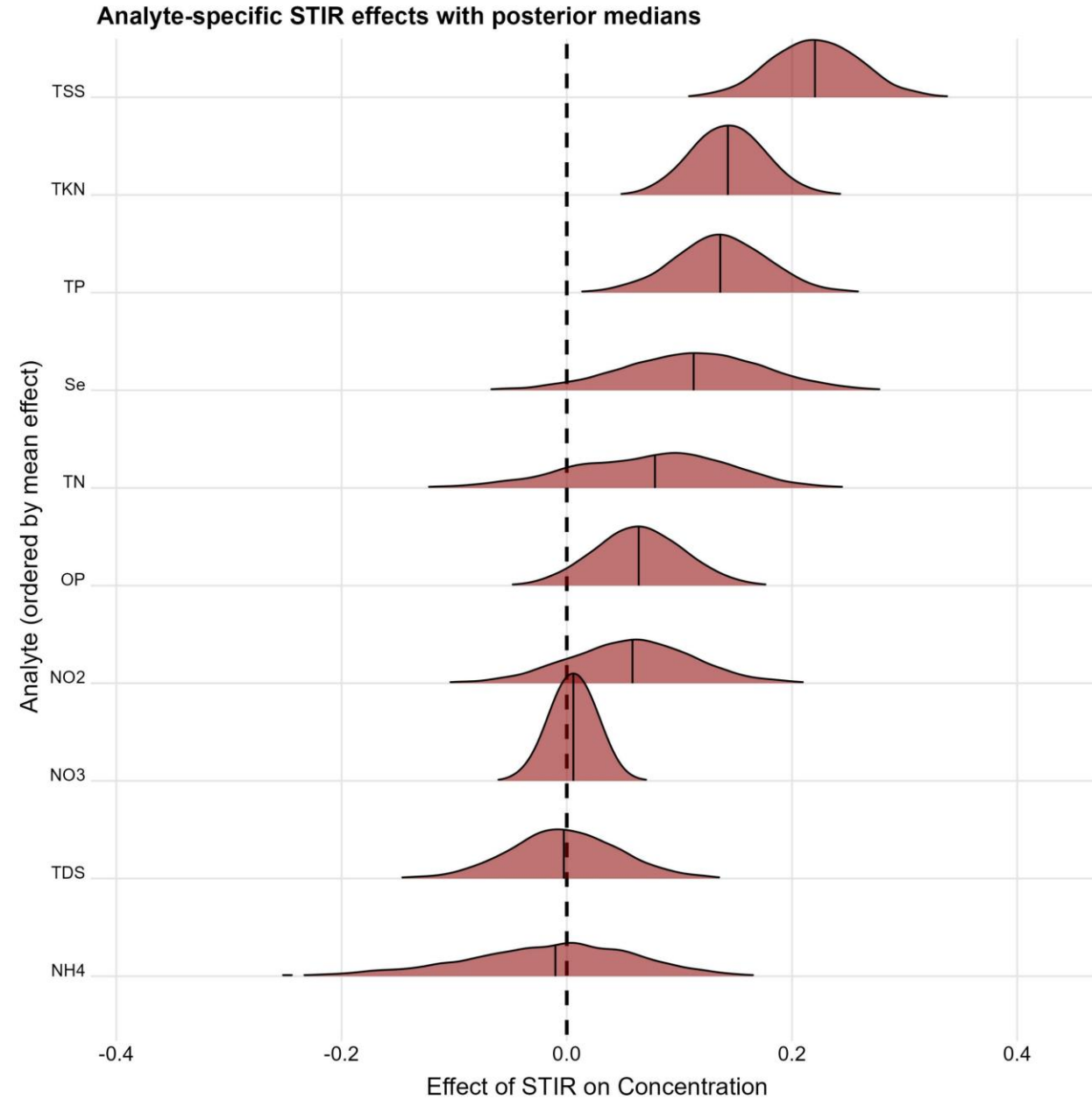
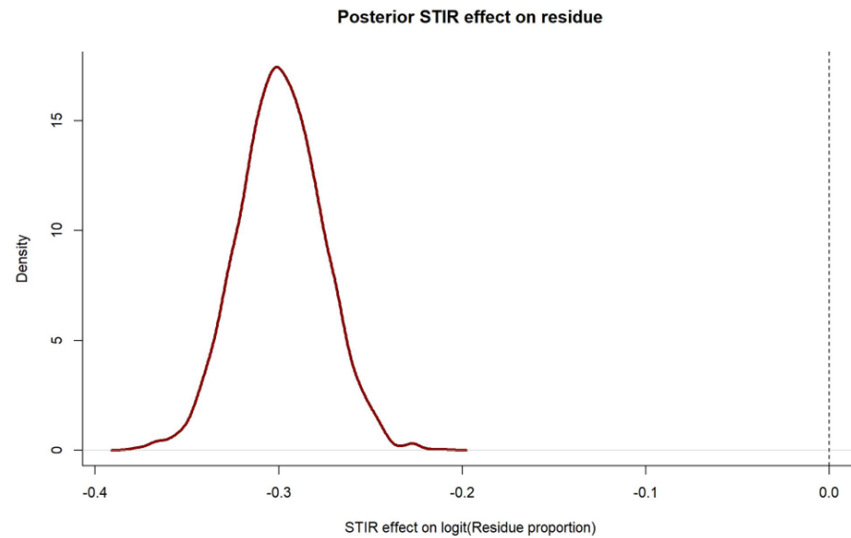
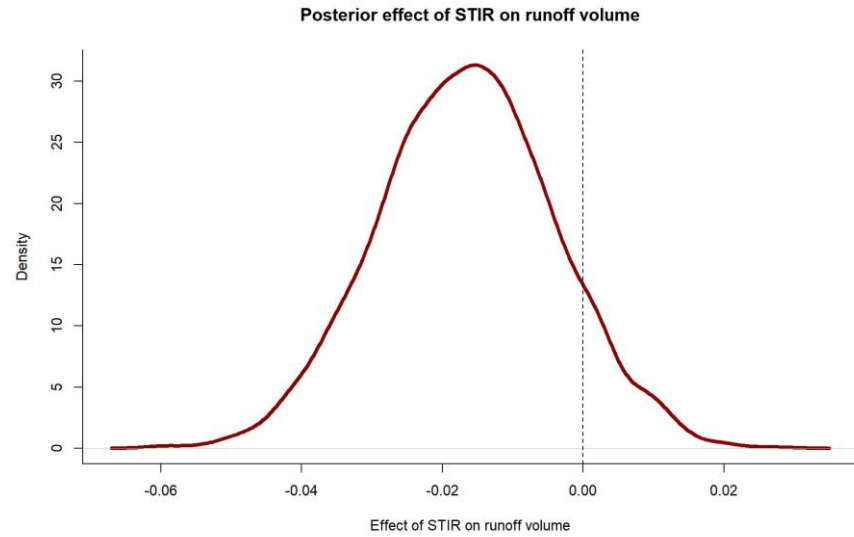


Irrigation Count Effects (i.e., First Flush)



- Minus TKN, all analytes tend to have “first flush” effects
- TSS, Se, and OP have the strongest effects
- TKN increases over each irrigation (decomposition??)

STIR effects



STIR impacts: ReLOADed

- Each panel shows the modeled relationship between Seasonal STIR and event load (g per irrigation) for one analyte.
- Points are observed events.
- Solid line is the posterior mean relationship.
- Dark ribbon shows uncertainty in the mean relationship.
 - *Answers: How certain are we about the average STIR-load relationship?*
- Light ribbon shows posterior predictive uncertainty.
 - *Answers: Given natural variability and measurement error, if we observed a new irrigation event at this STIR level, what range of loads should we expect?*

STIR → Load curves by analyte

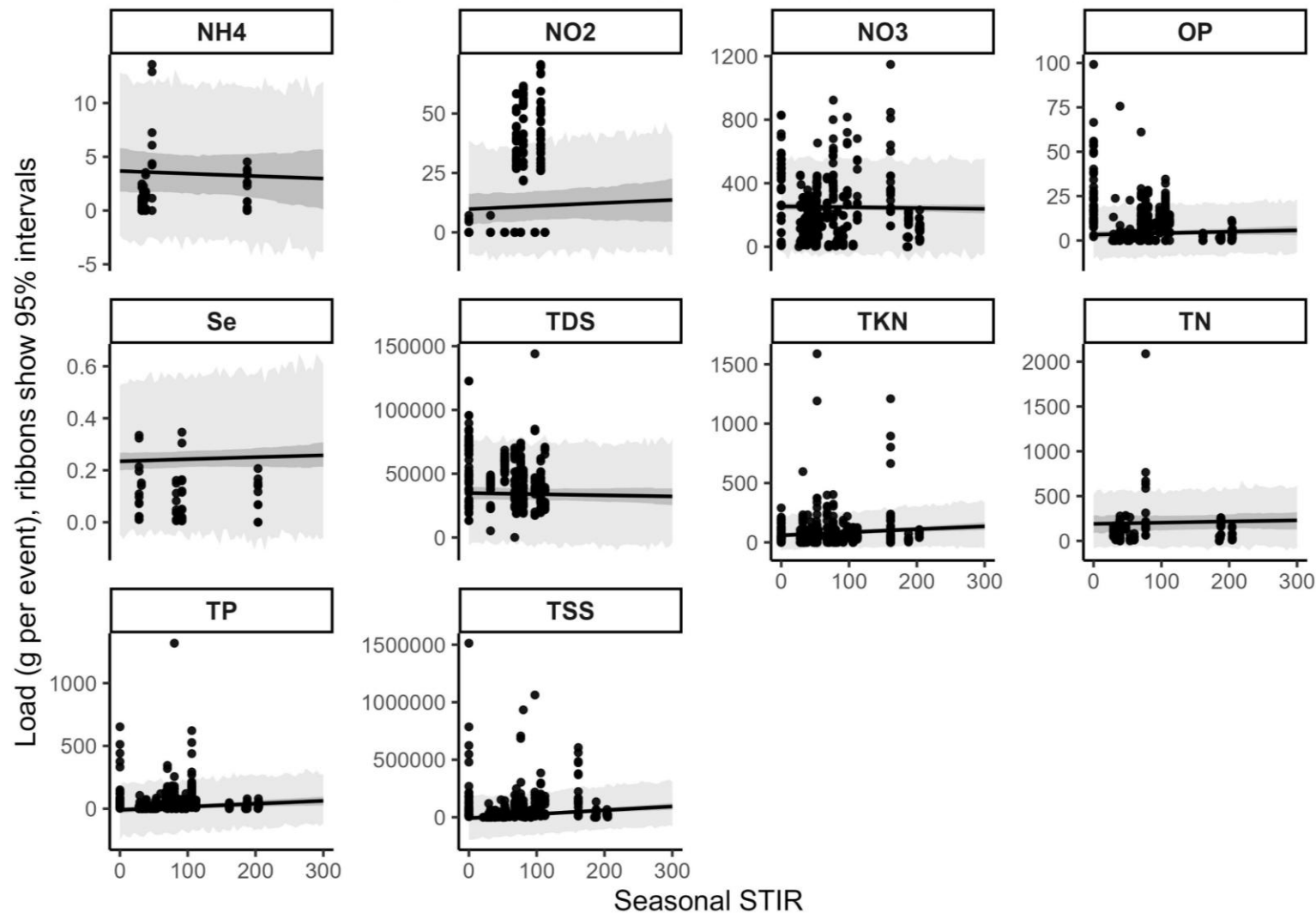
Year: avg_all_years; Block: avg_all_blocks; Sampler: avg_all_samplers;

Flume: avg_all_flumes; Irrigation: avg_all_irrigations;

Duplicate: avg_duplicates; Crop: avg_over_crops_weighted;

Residue: fixed 10.0%; Inflow(z)=0.00

Dark ribbon: mean relationship uncertainty. Light ribbon: posterior predictive (process + observation).



STIR → Load curves by analyte

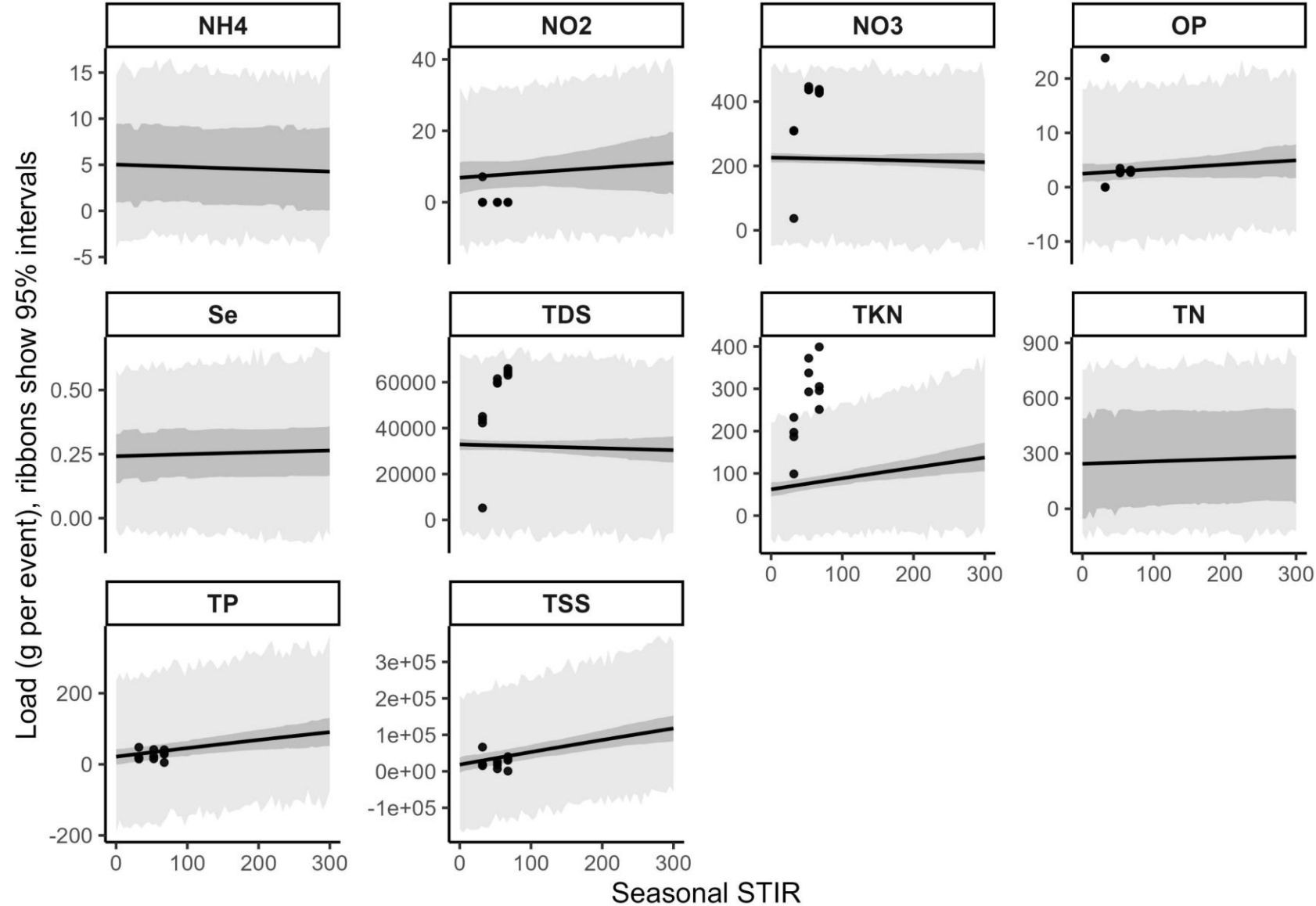
Year: 2023; Block: 1; Sampler: avg_all_samplers;

Flume: 60 V Trap; Irrigation: 1;

Duplicate: FALSE; Crop: silage corn;

Residue: fixed 15.0%; Inflow(z)=0.00

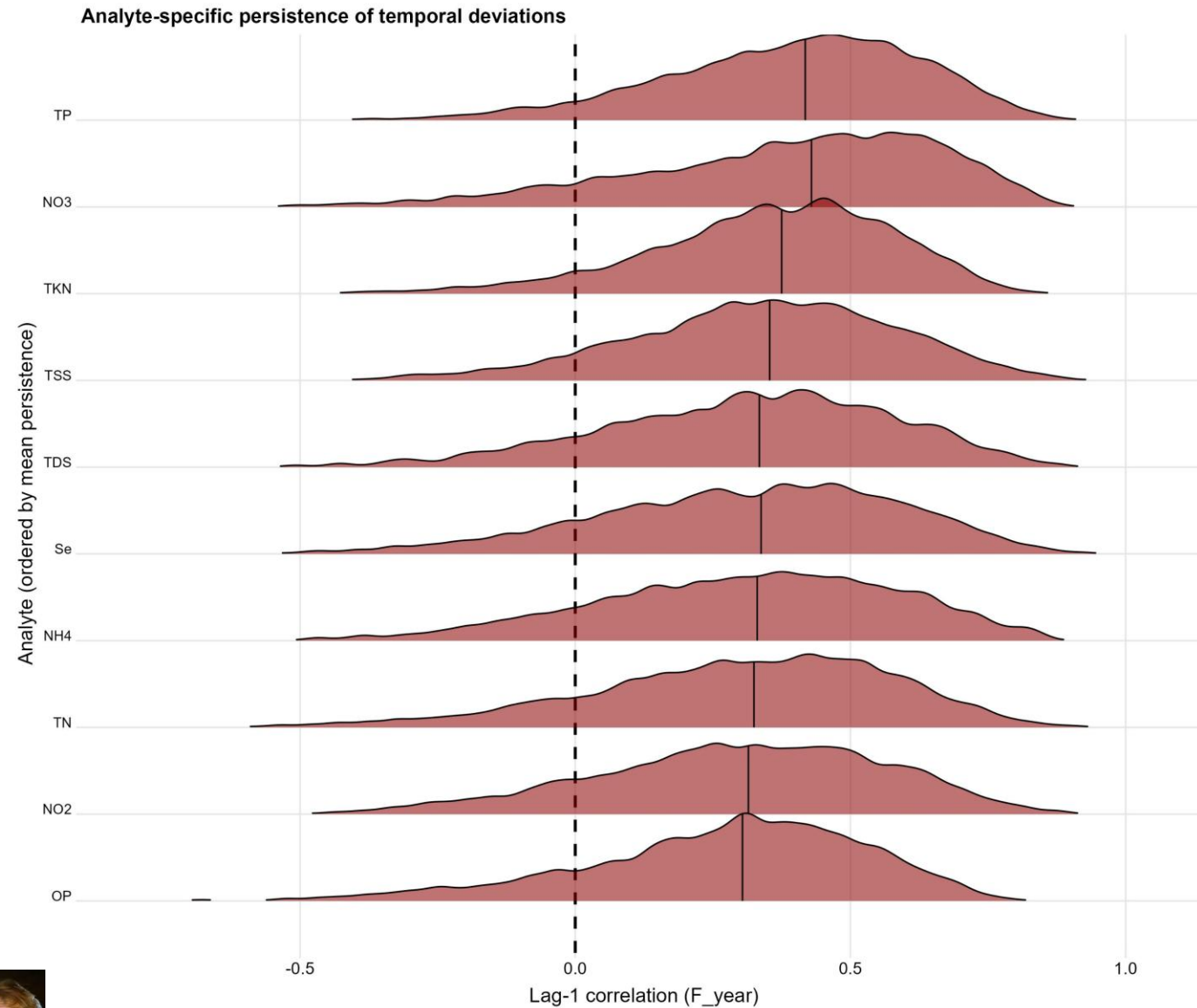
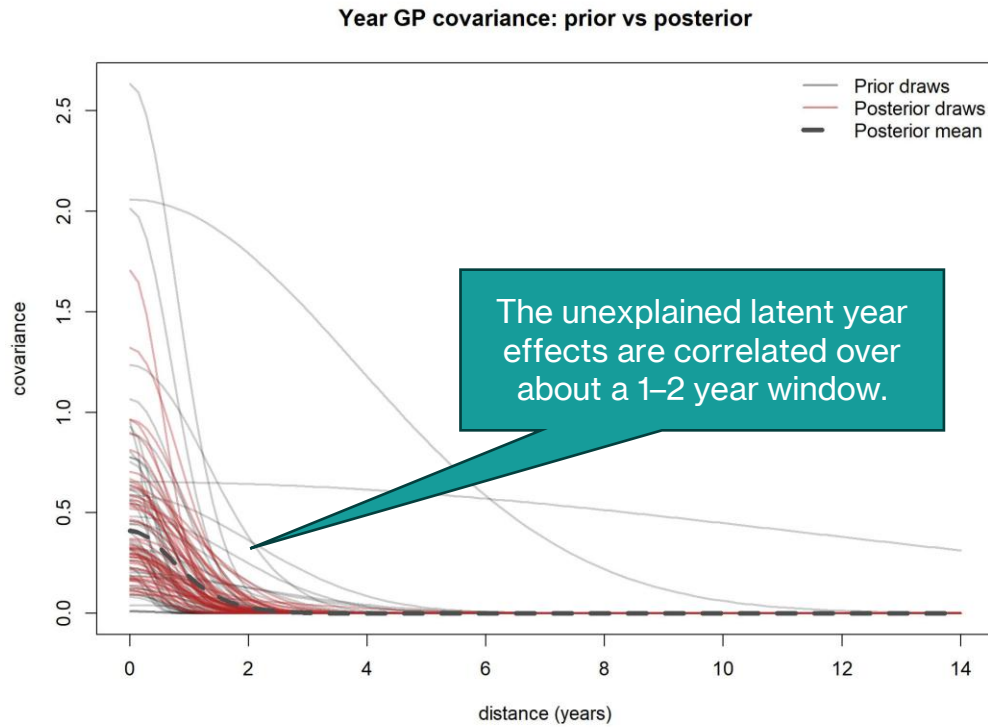
Dark ribbon: mean relationship uncertainty. Light ribbon: posterior predictive (process + observation).



Case example for better relationship viewing

- 2023 comparison study
irrigation 1

Back to the Future (Persistence effects)

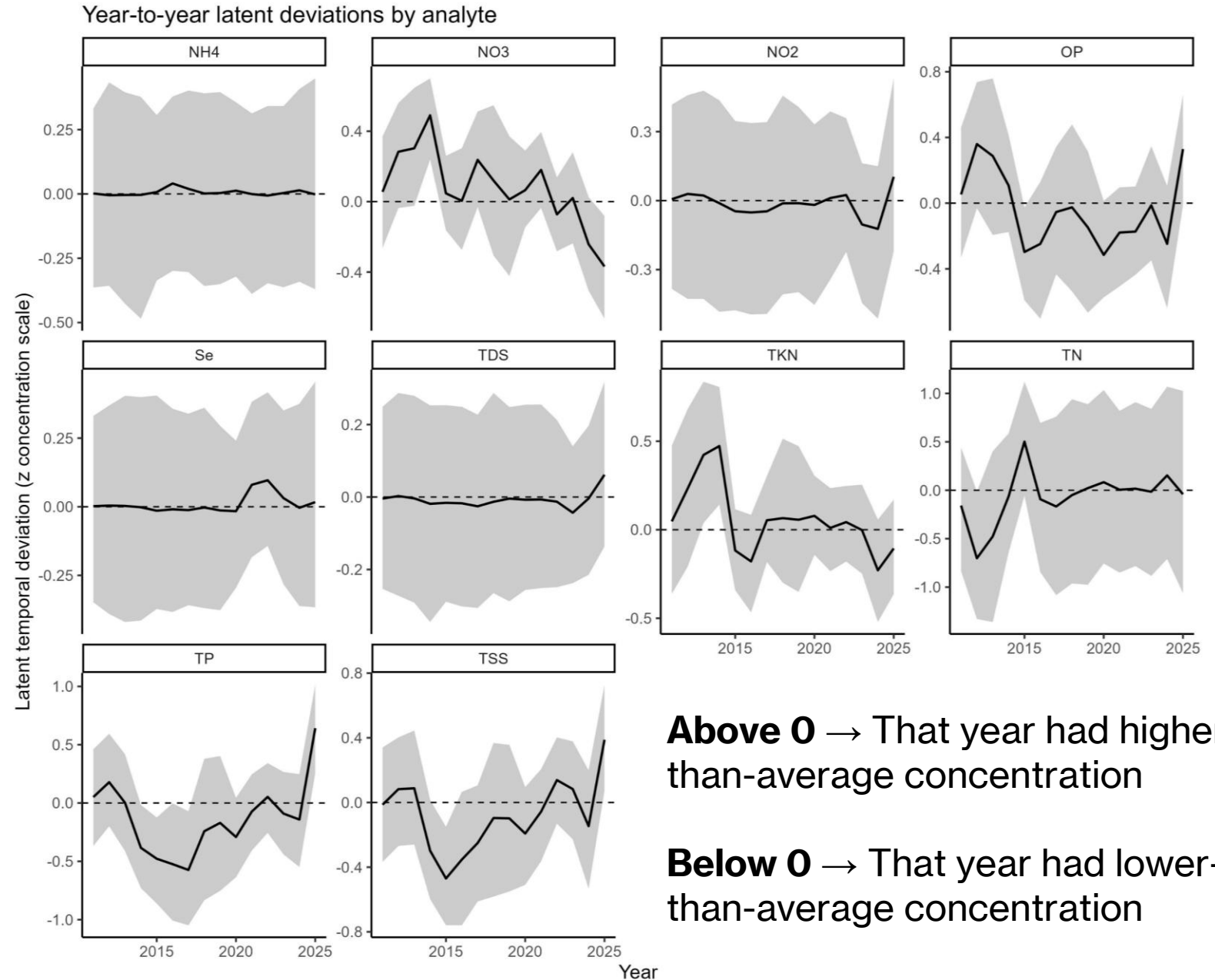


Correlation between year t and year $t+1$ for the latent year effect.

- Positive values mean adjacent years tend to move in the same direction.
- Negative values would mean oscillation or reversal.

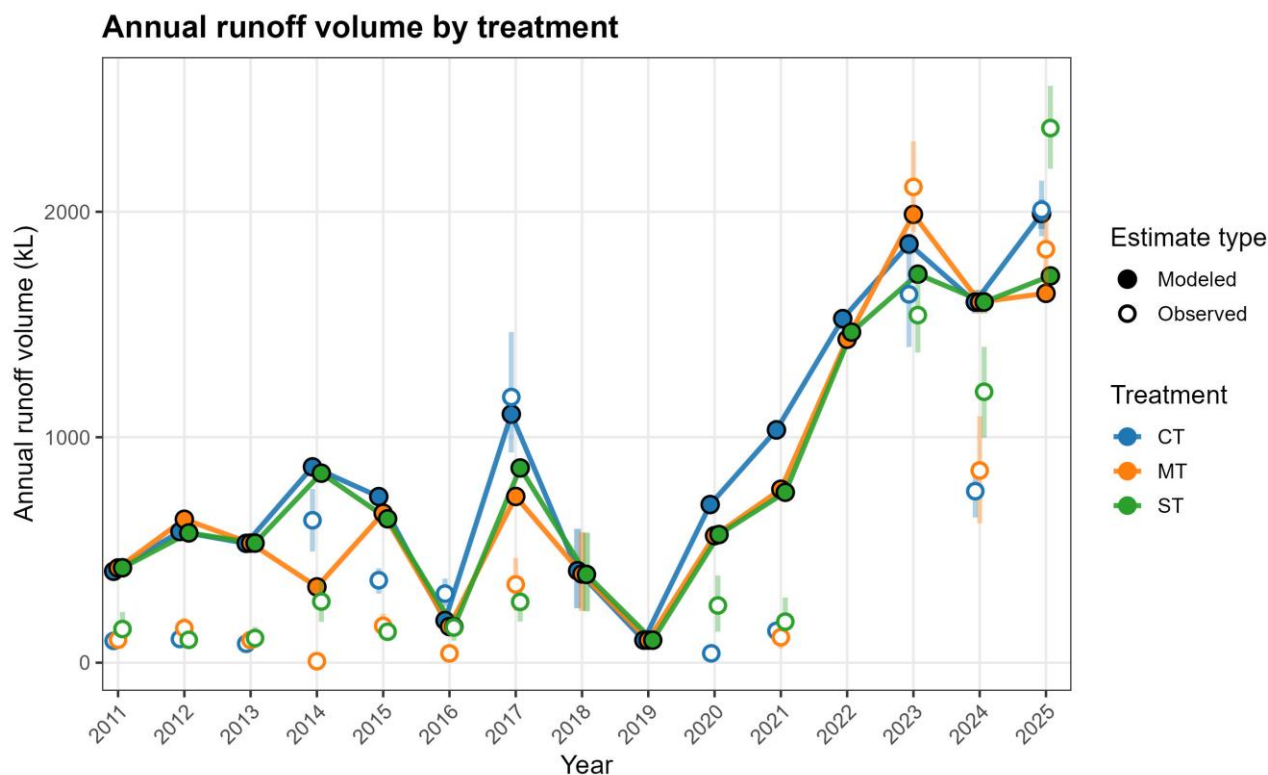
Back to the Future 2

- These lines show how each year deviates from the long-term average concentration **after accounting for management and measurement effects.**
- Values are on the z-scale of concentration.
- The dashed horizontal line at 0 represents the long-term average level for that analyte.
- The solid black line and 95% CI shading is the posterior mean year effect.



Annual Volumes

MT/ST had less runoff volume than CT overall
9 – 12% reduction



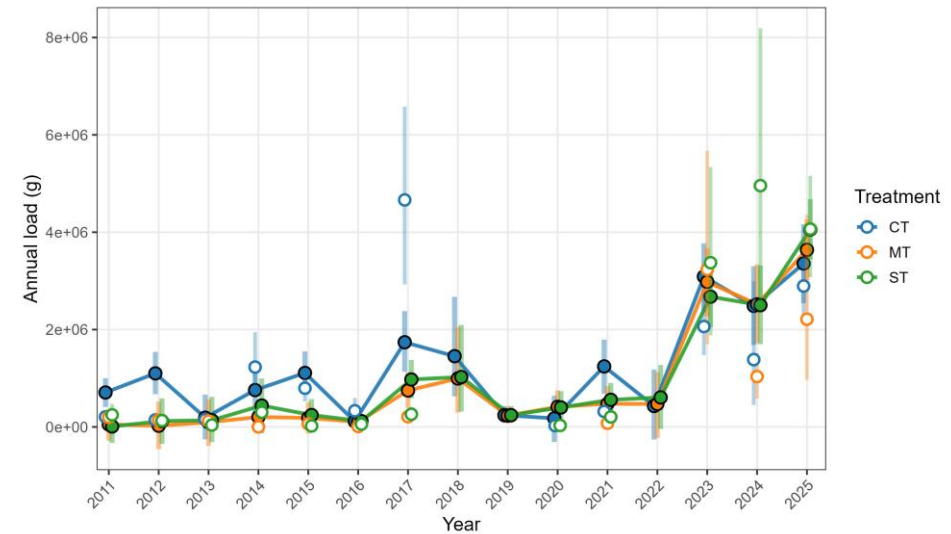
Modeled Annual Runoff Volume by Treatment Kiloliters (kL)

Year	CT	MT	ST
2011	406 (388, 424)	420 (404, 436)	421 (404, 437)
2012	582 (557, 607)	636 (613, 659)	576 (555, 596)
2013	529 (510, 548)	530 (511, 550)	530 (511, 550)
2014	869 (838, 896)	336 (323, 349)	840 (808, 872)
2015	737 (707, 766)	663 (644, 682)	638 (619, 658)
2016	187 (167, 208)	160 (141, 181)	159 (139, 179)
2017	1102 (1058, 1148)	737 (713, 762)	864 (832, 895)
2018	408 (242, 594)	394 (230, 578)	392 (228, 577)
2019	100 (97, 103)	100 (97, 103)	100 (97, 103)
2020	701 (674, 728)	563 (546, 582)	569 (550, 588)
2021	1033 (989, 1077)	769 (733, 804)	756 (727, 784)
2022	1526 (1477, 1570)	1435 (1401, 1469)	1466 (1430, 1501)
2023	1857 (1810, 1904)	1988 (1944, 2031)	1723 (1688, 1760)
2024	1600 (1549, 1654)	1600 (1549, 1654)	1600 (1549, 1654)
2025	1992 (1924, 2055)	1638 (1595, 1683)	1716 (1675, 1754)
Total	13627 (12986, 14282)	11971 (11443, 12522)	12349 (11813, 12906)

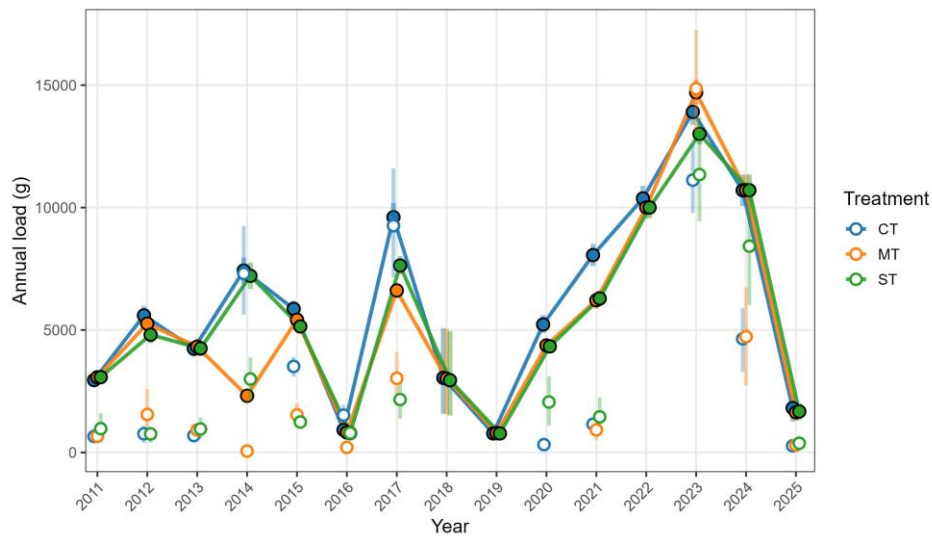
Annual Loadings

- Even though the model doesn't utilize treatment (CT/ST/MT), I thought it would be useful to calculate loads on a trt-level
- Graph shows modeled and observed data with 95% CIs
- This was done for all analytes
- Model does well on variables that have stronger STIR relationships!

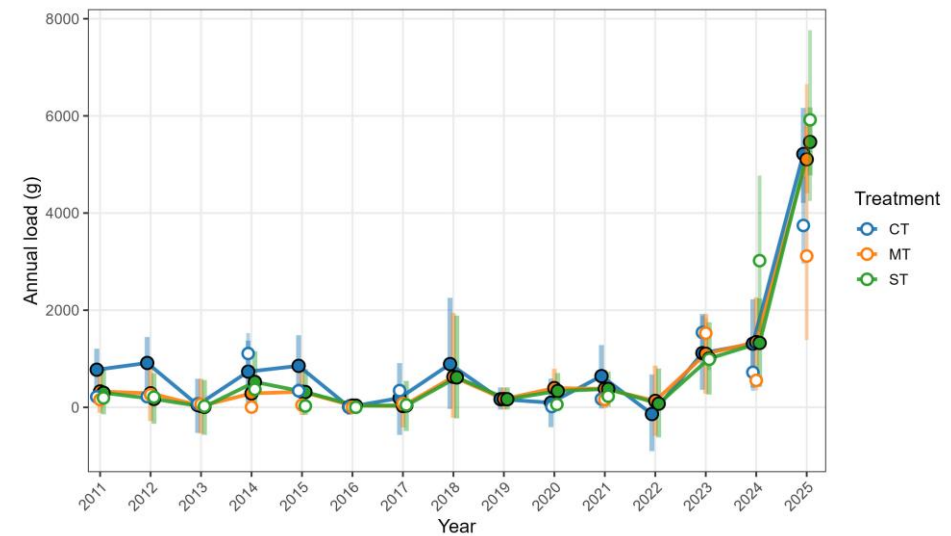
TSS: annual load by treatment



NO3: annual load by treatment



TP: annual load by treatment



What are the real impacts of CT/MT/ST?

Mass Loadings for Each Analyte by Treatment (kg) Over All Study Years

Analyte	CT	MT	ST	Mean % Reduction (MT, ST)
NH4	1.4 (0.1, 2.7)	1.2 (0.0, 2.3)	1.2 (0.0, 2.4)	(16%, 11%)
NO3	90 (83, 98)	79 (73, 86)	82 (76, 90)	(12%, 9%)
NO2	4.2 (1.1, 6.9)	3.1 (0.7, 5.3)	3.4 (0.7, 5.8)	(26%, 19%)
OP	1.9 (1.1, 2.7)	1.8 (1.1, 2.5)	1.8 (1.1, 2.6)	(6%, 3%)
Se	0.09 (0.07, 0.12)	0.08 (0.06, 0.10)	0.08 (0.06, 0.10)	(18%, 15%)
TDS	12,863 (10,842, 14,921)	11,520 (9,804, 13,212)	11,845 (10,031, 13,656)	(10%, 8%)
TKN	26.0 (17.7, 34.7)	18.9 (11.7, 26.4)	20.2 (12.8, 28.0)	(27%, 22%)
TN	79.24 (5.60, 153.56)	71.44 (5.40, 141.97)	73.64 (6.96, 145.2)	(10%, 7%)
TP	12.8 (3.4, 22.7)	10.6 (2.6, 19.1)	10.8 (2.6, 19.4)	(18%, 16%)
TSS	18,197 (10,234, 26,641)	13,138 (6,332, 20,260)	14,107 (7,230, 21,455)	(28%, 22%)

- On average, conservation tillage reduced loadings across the board!
- Most significant impacts on sediment bound analytes
- No real differences between ST and MT
- Uncertainty is wide but honest to real-world variability
- Both MT and ST strategies improve WQ, so chose what's easier to adopt

A note on TN

	CT	MT	ST
TN Modeled (kg)	79.2 (5.6, 153.6)	71.4 (5.4, 142)	73.6 (7, 145.2)
TN Calculated (kg)	120.7 (102, 139.9)	101.2 (85.5, 117.8)	106.2 (89.8, 123.4)

Previous slide showed modeled TN, but summing the means of its modeled components (NO₃, NO₂, TKN) yielded higher than purely modeled forecasted mean.

However, note that the calculated TN (i.e., summing modeled component means) falls inside the purely modeled TN, indicating that the model is doing well at capturing that potential uncertainty in lieu of not understanding mechanistic processes.

Summary and Discussion

- The Bayes framework allowed us to look at data causally, even in adverse data conditions, and should be a model for other researchers
- The model appears to have captured **inference** well, and **prediction** for most parameters
- ST/MT perform very similarly in aspects of mass loading and runoff volume reduction
- CT was consistently worse for WQ with more runoff compared to ST/MT

- Questions
 - With so much potential insight, what are the key things the paper needs to say?
 - AJ ideas: 1) STIR impacts on WQ, Vol, and Loads, 2) Temporal persistence, 3) practical implications (load reduction potential and adoption considerations)
 - Who should contribute to the paper, and in what capacities, given this is a long-term AWQP effort but also that it's AJ's dissertation?
 - Current authors: Ansley J. Brown ^{a,1}, Emmanuel Deleon ^a, Erik Wardle ^a, Troy Bauder ^a, Jakob F. Ladow ^a, Tad Trimarco ^b, Lauryn Howlett ^a, Allan A. Andales ^a - **Any others??**

Intermission

Switching gears to PhD Ch3: Bayes Vs. ML
(In Progress)

**Understanding the Application of Bayesian and Machine Learning Models
for Environmental Data: A Long-Term Runoff Water Quality Case Study**

Ansley J. Brown, Tad Trimarco, Allan A. Andales, Emmanuel Deleon, Erik Wardle, Troy
Bauder, Jakob F. Ladow



Ch3: Rise of the Machine (Learning)

- Environmental data are messy!
 - E.g., Non-normal, missing, extreme values, seasonality, complex interactions and confounding
- Bayesian and machine-learning (ML) approaches are becoming popular
- Despite this, there is limited guidance on how their differing assumptions and uncertainty formulations influence their usage for environmental timeseries data
- **Objective:** to provide a comparison between a Bayesian and machine-learning (ML) model calibrated with the same agronomic dataset to provide clarity on when each approach is more appropriate for a user's needs.

A Tale of Two Models

BAYESIAN APPROACH

- Hierarchical GLMM – linear model
- Explicit causal framework (DAG based)
- Models parameter, process, and measurement uncertainty explicitly
- Imputes missing data
- Probability based, with stated assumptions
- Implemented in Stan via R and C++

MACHINE LEARNING

- Gradient boosted decision trees (CatBoost)
- Pattern learning from data, minimal assumptions
- Captures nonlinear effects and interactions automatically
- Uncertainty learned from conformal CV process
- Built in Python

The (Model) Hunger Games

Because Bayesian and ML approaches answer different questions, we evaluate them using both quantitative and qualitative criteria.

- **Quantitative** – how well does each model perform at data characterization?
- **Qualitative** – How well does each model explain and support decisions?



Quantitative Metrics of Comparison



Data visualizations, comparing annual load predictions and uncertainty bands for each model



Normalized root mean squared error (NRMSE), used to assess point-prediction accuracy at the event scale under cross-validation, reflecting common practice in environmental prediction studies.



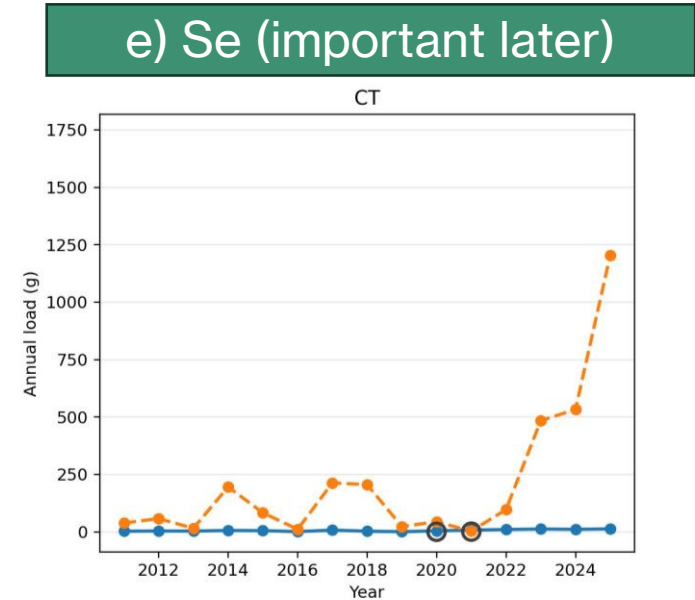
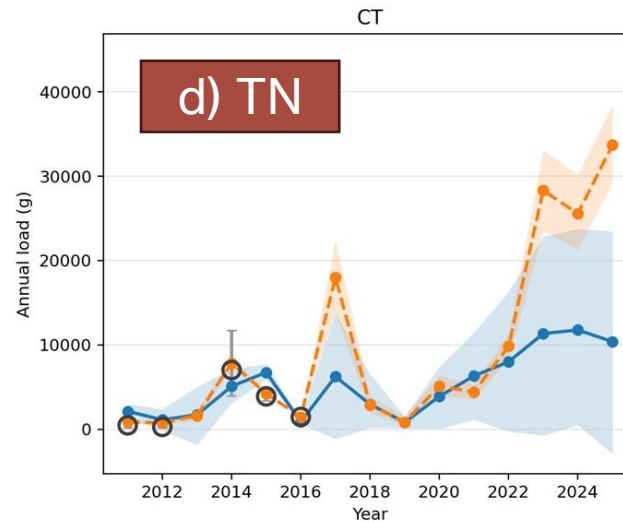
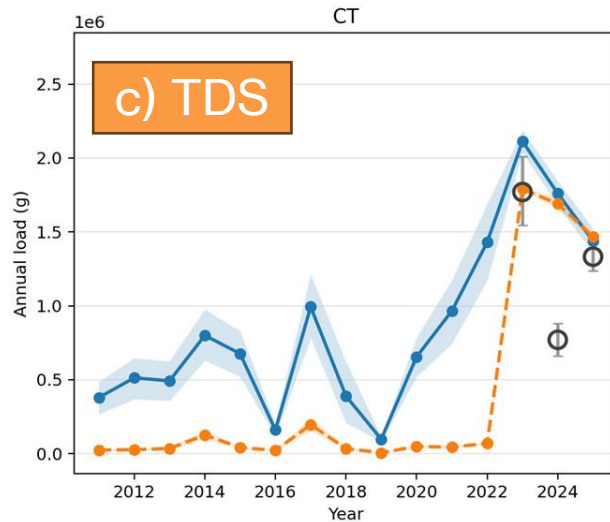
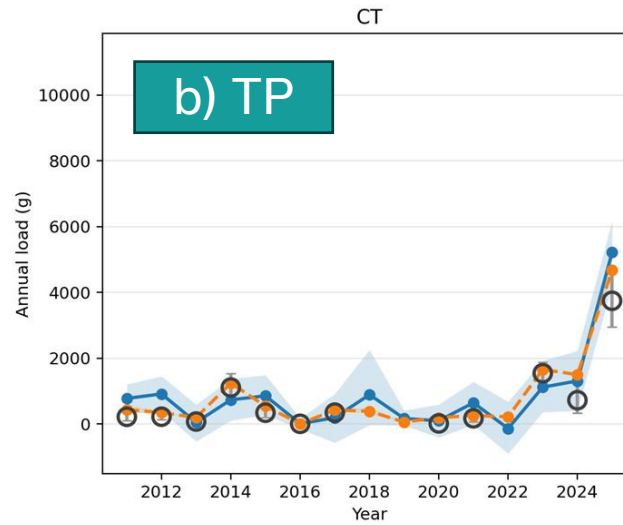
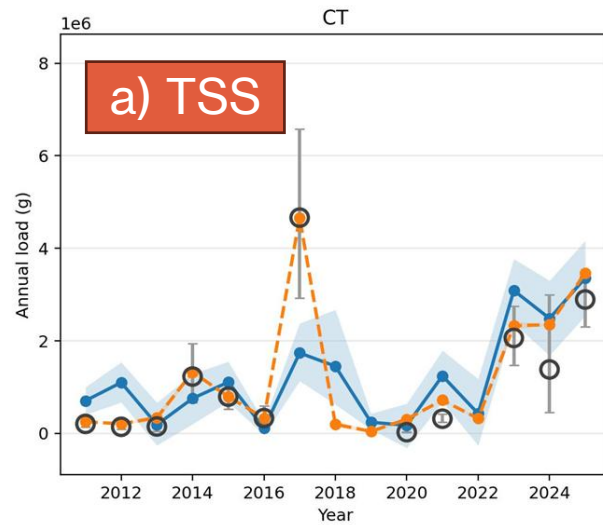
Normalized continuous ranked probability score (NCRPS), used to evaluate the quality of probabilistic predictions by jointly assessing accuracy and sharpness of predictive distributions.



Empirical coverage of nominal prediction intervals, used to assess calibration of model-based uncertainty estimates.

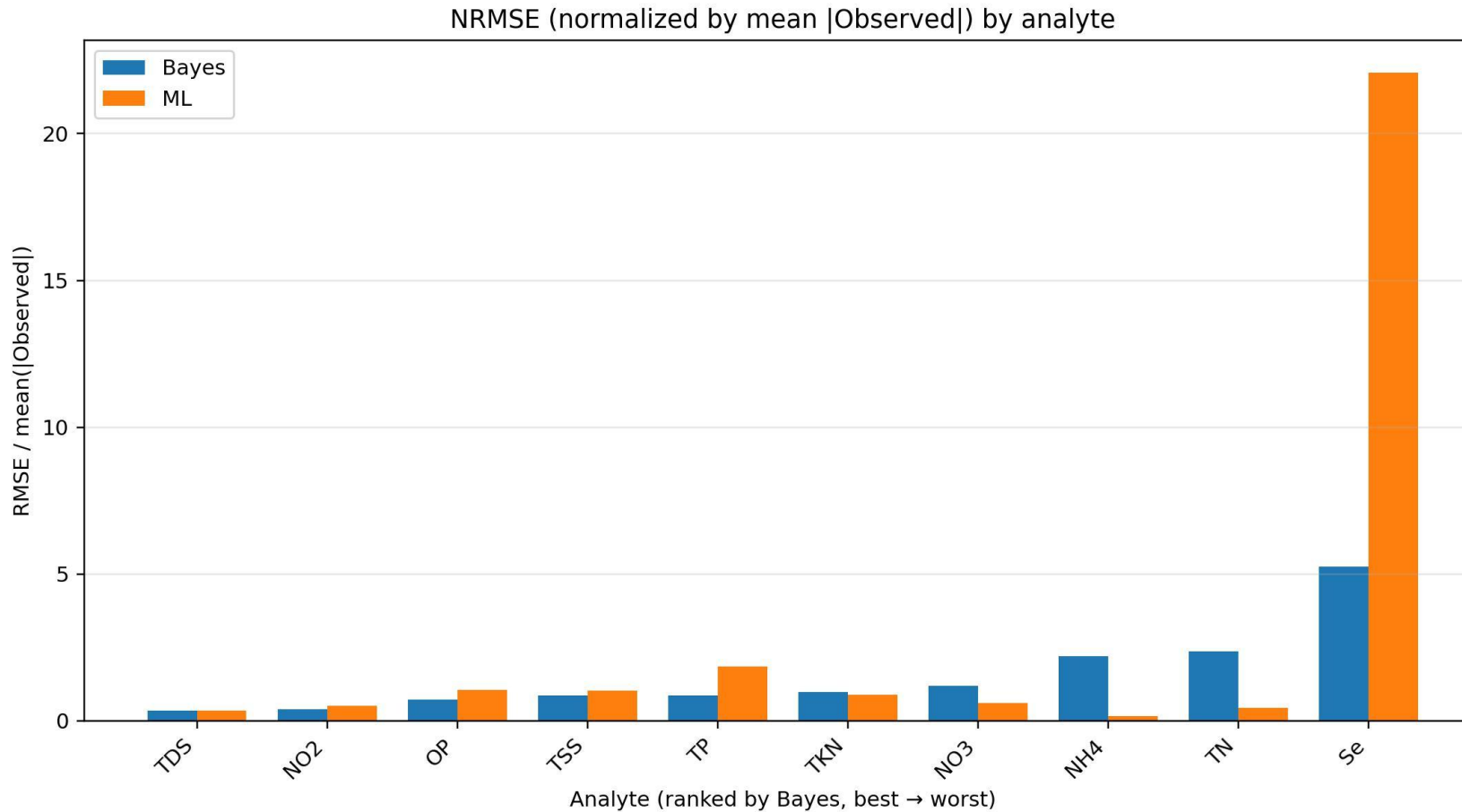
Visual Results

- Both models do a similar job at predicting observed outcomes



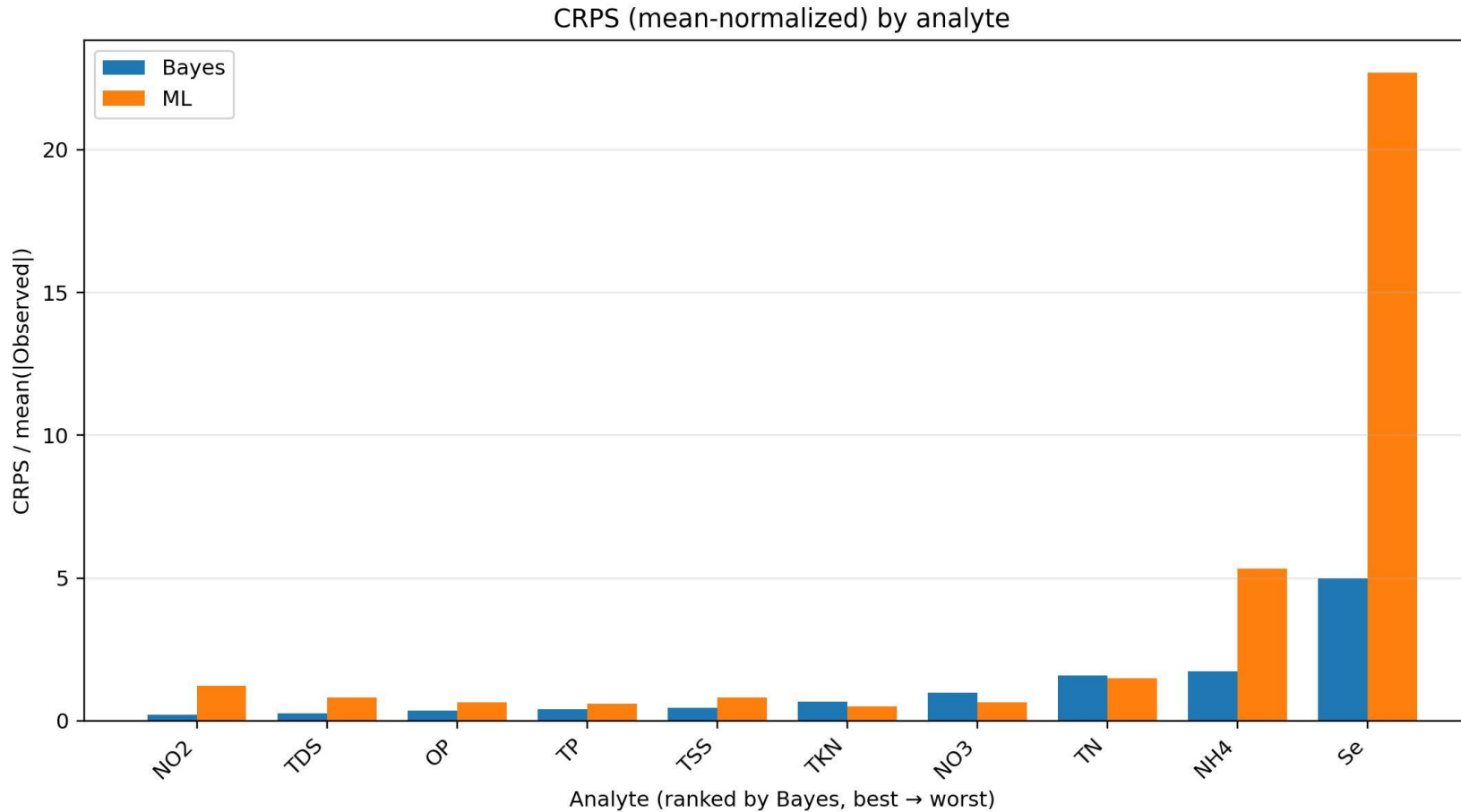
NRMSE

- Prediction point performance metric (lower = better)
- ML is competitive and even better for some analytes like TKN, NO3, NH4 and TN, but fails dramatically for Se.



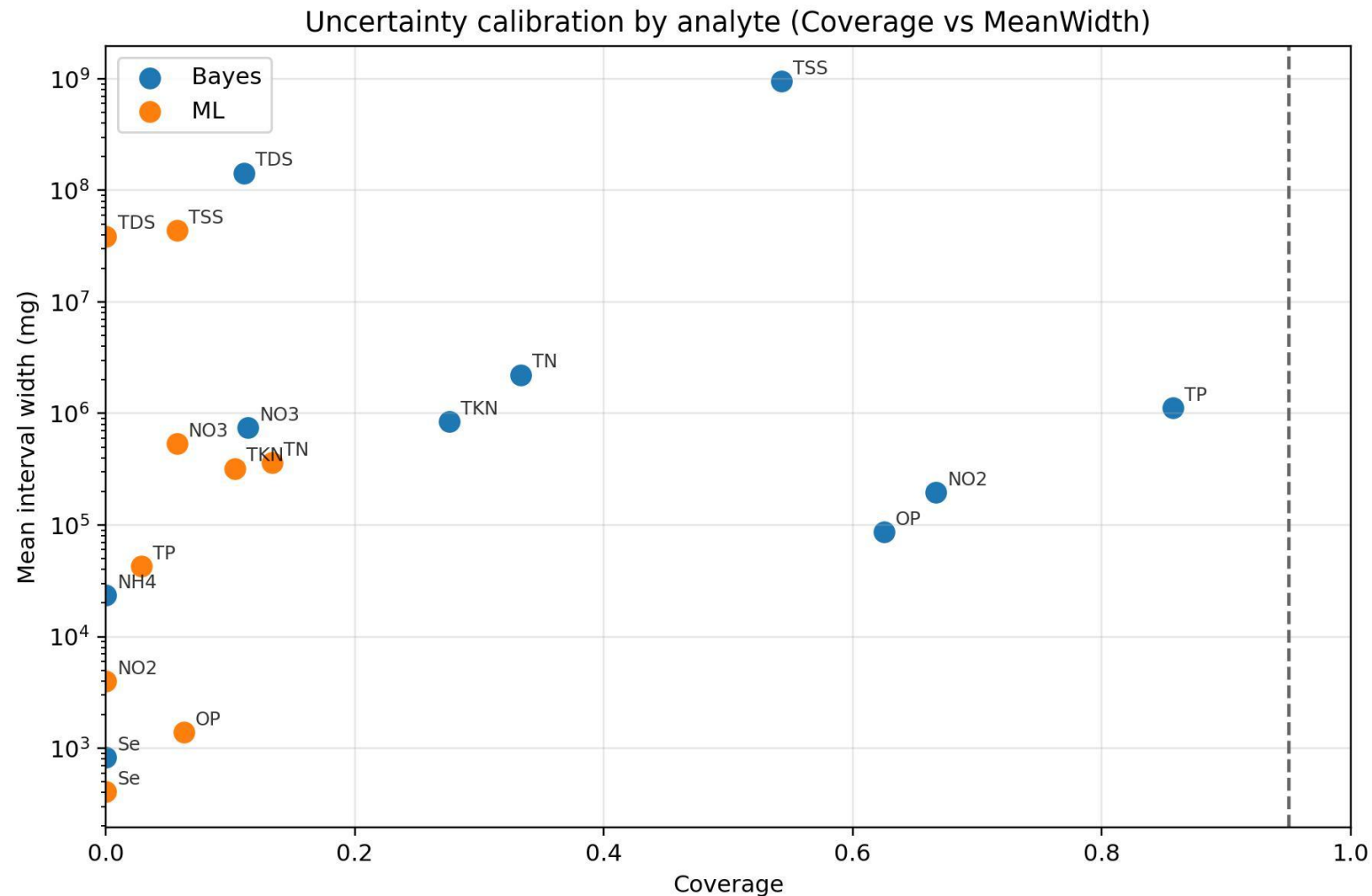
NCRPS

- This compares full predictive distributions. Lower means the entire distribution better aligns with observed values.
- Bayes outperforms ML for most analytes, particularly Se and NH₄, indicating better probabilistic calibration.



Coverage

- This shows whether uncertainty intervals are correctly calibrated.
- ML intervals are much too narrow and severely under-cover.
- Bayes intervals are wider and better calibrated but still do not achieve nominal 95% coverage.



Qualitative Themes of Comparison



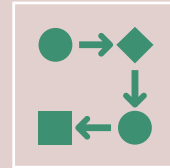
Uncertainty and robustness:
How uncertainty is defined
and behaves under sparse or
noisy data



Causal interpretability:
Ability to estimate
management effects and
support defensible decisions



Generalization: Forecasting
future years and applicability
beyond the training regime



Practical implementation:
Complexity, runtime, and
workflow fit for real-world
application

Key Differences: Missing Data

BAYESIAN APPROACH

- Missing data are unknown parameters.
- The model jointly estimates missing values with observed data.
- Missingness affects parameter estimates directly.
- Imputation is model-based and conditional on model parameterization made by user

MACHINE LEARNING

- Missing target values are excluded from training.
- No explicit modeling of unobserved events.
- Predictions are made only for observed event structure.
- Missing events are not modeled unless explicitly engineered (as we did here).

ML uncertainty is predictive, not structural.

Key Differences: Uncertainty

BAYESIAN APPROACH

- Uncertainty is part of the generative model.
- All outputs have probability distributions.
- Uncertainty propagates through the hierarchy.
- Annual load reflects uncertainty from:
 - Parameter estimation
 - Process variability
 - Measurement error
 - Missing data
 - Cross-analyte and temporal structure
- Credible intervals represent posterior probability.

MACHINE LEARNING

- Uncertainty is added after model training.
- The model produces point predictions.
- Conformal prediction wraps predictions with coverage intervals (not distributions).
- Uncertainty comes from:
 - Prediction residuals
 - Split conformal calibration
 - Bootstrap aggregation
 - Prediction intervals guarantee coverage, not generative structure.
- Intervals guarantee coverage, not structural interpretation.

Bayesian uncertainty is structural and model-based.
ML uncertainty is distribution-free and predictive.

Qualitative Metric Results

Evaluation Criteria	Bayesian Modeling	Machine Learning (CatBoost Gradient Boosting)
Uncertainty handling approach	Parameter, measurement, and process uncertainty modeled explicitly within a probabilistic generative framework (via HMC)	Prediction intervals derived from split conformal calibration; not parameter-based
Uncertainty interval interpretation	Full posterior distributions for parameters and predictions given model and data	Prediction interval calibrated to achieve target coverage for future observations.
Missing data handling	Models missing values as latent variables and propagates their associated uncertainty	Requires observed outcomes; missing values handled via preprocessing or exclusion
Measurement error handling	Explicitly characterizes measurement error and natural variability vs. effect (i.e., parameter) uncertainty	Treats measurement method as a model feature and “learns” patterns if signal exists; does not distinguish error types
Interpretability of model parameters	Yes, built around a structured model to isolate direct effects	No, direct causal effect parameters; feature importance does not imply intervention effect
Interpretability of model output	Generates joint predictions of concentration, volume, and loads with coherent uncertainty	Predicts concentration and volume separately; load uncertainty propagated via simulation
Forecasting ability (i.e., future trends at the same site)	Uses explicit time structure, uncertainty widens farther from data	Performance evaluated empirically via held-out years; extrapolation outside training regime is not structurally constrained.
Ease of Implementation	More complex, requires user expertise and causal assumptions	Easier to implement, tool explores data on its own
Computational Burden (current config.)	5.5-hour runtime	2-hour runtime
Transferability (i.e., model value when applying to new sites)	Causal and statistical framework generalizes conceptually; direct transfer limited without refitting	Model parameters are highly training data specific; performance depends on similarity of new site data
Behavior under sparse and/or unreliable data	Partial pooling stabilizes estimates under low sample size; uncertainty widens appropriately; extreme values moderated by hierarchical structure	Performance degrades with limited and/or unreliable data; may produce unstable or overly confident predictions in sparse regimes without additional constraints (e.g., Se).
Regulatory, scientific, or conservation planning defensibility	Model assumptions, priors, and structure are explicit and auditable	Primarily data-quantity and output driven; less transparent mechanisms

Decision Framework

	Bayesian	Machine Learning
Best suited for...	• Inference • Long term system understanding • Decision making under uncertainty (e.g., to understand how tillage affects water quality and loads) • Interpretable intervention impacts	• High prediction accuracy • Nonlinear pattern detection • Empirical performance benchmarking • Rapid deployment with low application expertise

Example Decision Considerations:

- Do you need causal effect estimates? → Bayesian
- Do you need highest predictive accuracy? → ML
- Is missing data substantial? → Bayesian
- Do you have very few data? → Bayesian
- Are your data non-normal? → Both, but differently
- Are you forecasting future years? → Both, but differently
- Do stakeholders need interpretable effect sizes? → Bayesian



Summary

- Both approaches predicted annual loads reasonably well.
- The Bayesian model helps explain what is driving changes in water quality.
- The ML model focuses on making accurate predictions.
- Bayesian intervals show how confident we are in the system itself whereas ML intervals show how far off a prediction might be.
- ML has a low barrier to entry whereas Bayes relies heavily on the user's knowledge
- The two approaches serve different purposes and should be used in different applications



Hydrology Days?

- **Call for Abstracts**

Abstract submissions are now open:

<https://forms.office.com/r/tGCCib3nWS>

Abstract Deadline: March 24, 2026

- **Registration**

All presenters and attendees are required to register:

<https://forms.office.com/r/LDvNBKkJbZ>

Registration Deadline: March 31, 2026

Registration Fee: \$50